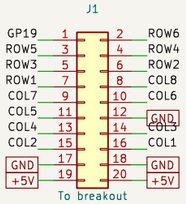
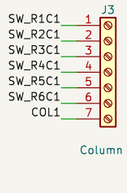
Breakout board 20-pin header (straight IDC ribbon, pin 1 to pin 1)

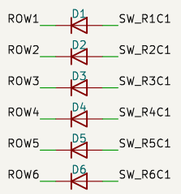


Switch matrix: one screw terminal and one diode per switch. Wire each switch between its terminal (1–6) and the column's COM terminal. Diode stripe (K) faces the row.

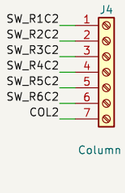
Rows: GP13–GP18 (breakout S2 A1 S1 L R D). Columns: GP5–GP12 (K4 K3 K2 K1 P4 P3 P2 P1).

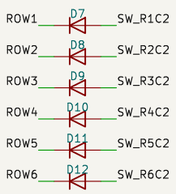
Column 1: GP5 (K4)



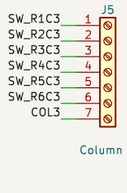


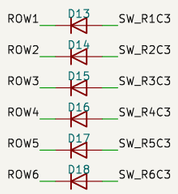
Column 2: GP6 (K3)



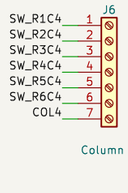


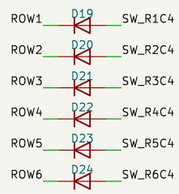
Column 3: GP7 (K2)



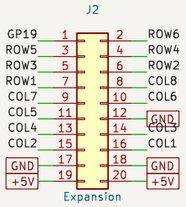


Column 4: GP8 (K1)

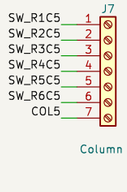


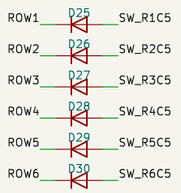


Expansion: same pinout, wired in parallel with J1

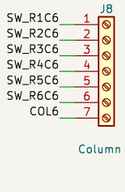


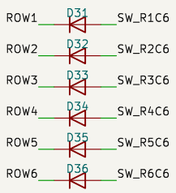
Column 5: GP9 (P4)



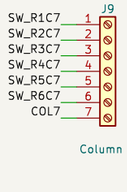


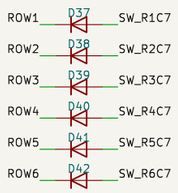
Column 6: GP10 (P3)



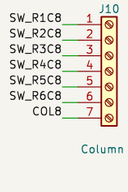


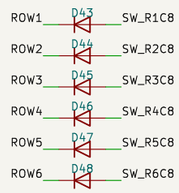
Column 7: GP11 (P2)





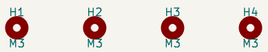
Column 8: GP12 (P1)





GPIO 19 stays a normal button (UP), wired to ground





D1–D48: 1N4148W (SOD-123), cathode/stripe to ROW

Firmware: RP2040AdvancedBreakoutBoardUSBPassthroughMatrix
For RP2040 Advanced Breakout Board – USB Passthrough, 20-pin header

Sheet: /
File: keymatrix-adapter.kicad_sch

Title: GP2040-CE Key Matrix Diode Adapter

Size: A3	Date:	**Rev: 1.0**
KiCad E.D.A. kicad-cli 7.0.11+dfsg-1build4		Id: 1/1